

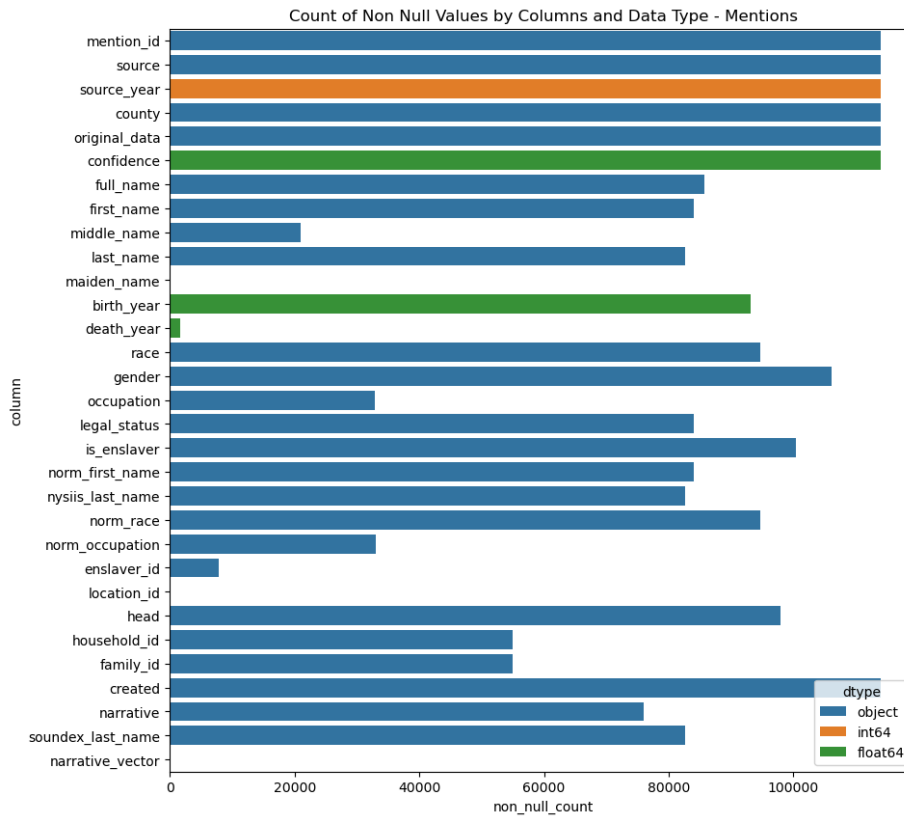
Visualization Report

Report contains data visualizations for the *mentions*, *assertions* (6/27/2026 iteration), *ground truth*, and *candidate matches* datasets.

Exploratory Data Analysis (EDA)

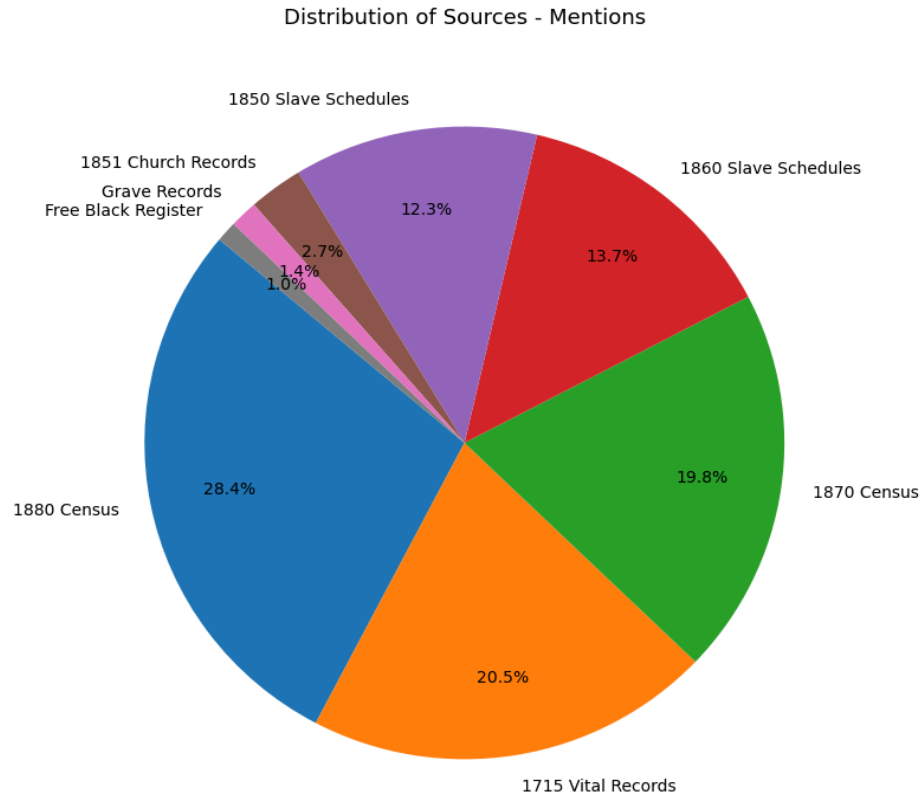
Mentions Data

Figure 1:



The mentions dataset has **114,070** observations and **31** features/columns. The 'maiden_name', 'location_id', and 'narrative_vector' features contain all NA values, while the 'mention_id', 'source', 'county', 'original_data', and 'confidence' features contain no NA values.

Figure 2:



*While the mentions dataset includes information from **8** source documents, the majority of this data comes from the 1870 and 1880 censuses, accounting for **~48.2%** of observations.*

Figure 3:

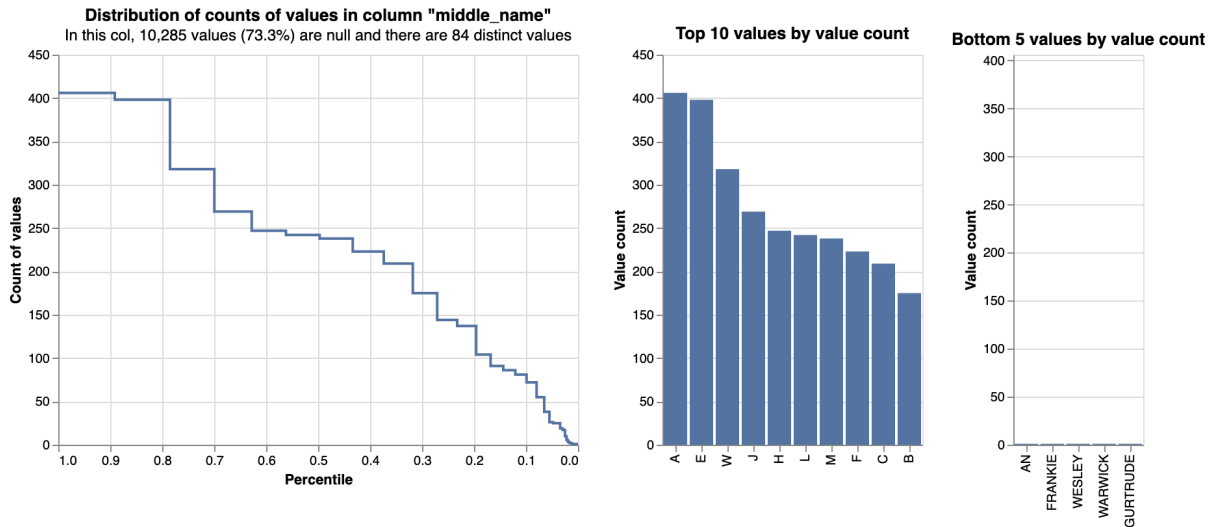
Last Names with Nysiis Code BRAN

last_name	count
Brown	2118
Braun	7
Bruns	6
Brim	6
Brown?	5
Browen	5
Brann	3
Brian	3
Brien	3
BROWN	2
Brunn	2
Brina	1

*This table shows the last names that are all encoded as **BRAN** in NYSIIS format. They will therefore be considered the same last name when compared in candidate matching.*

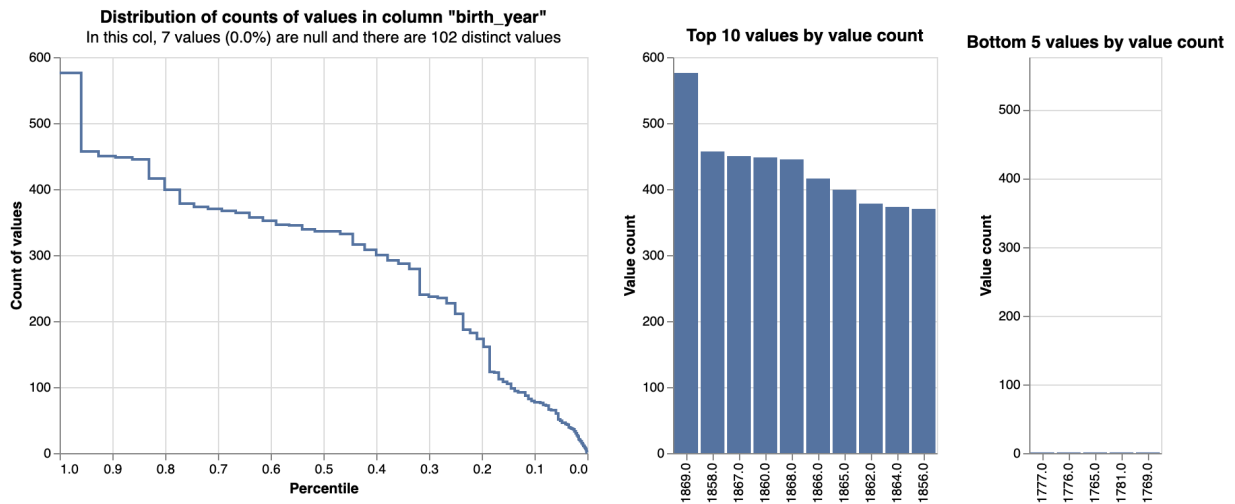
1870 Census Data

Figure 4:



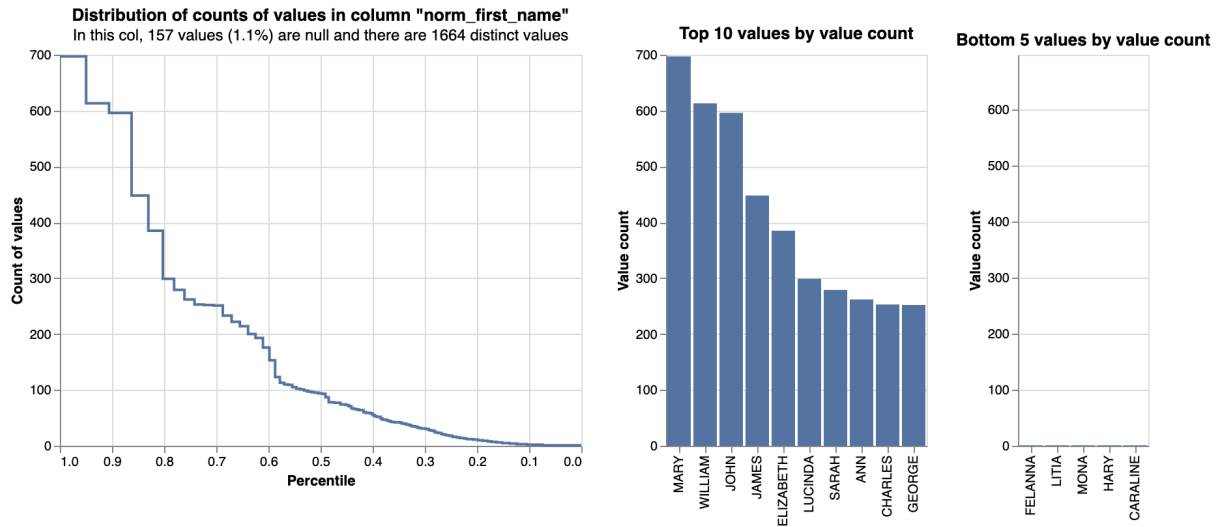
The combined figure depicts data from the 1870 Albemarle County Census data. The figure on the left depicts the distribution of count values for the variable 'middle name'. The two figures on the right depict the most common and least common entries for "middle name".

Figure 5:



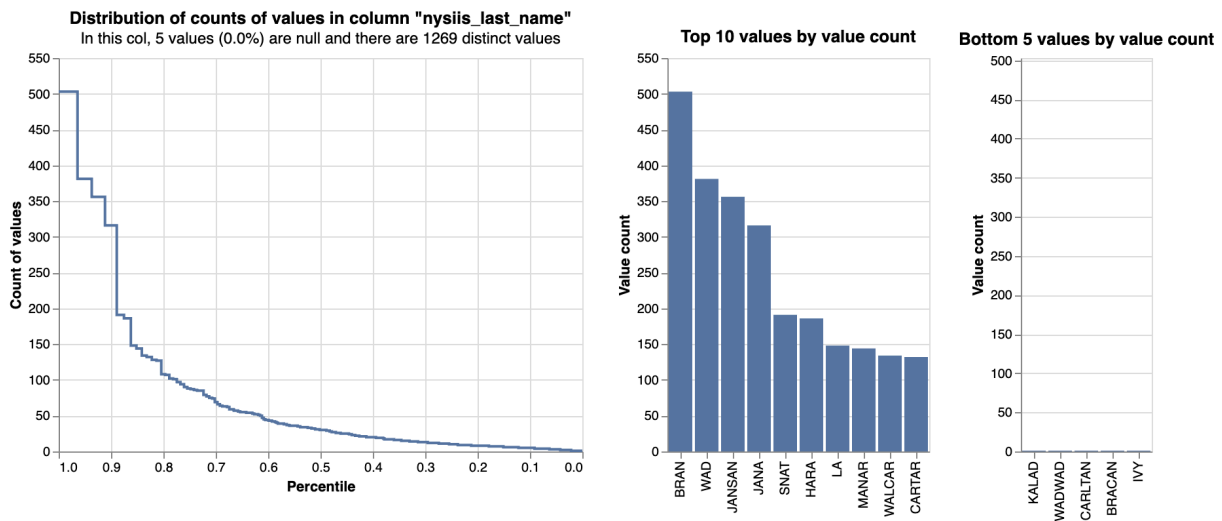
The combined figure depicts data from the 1870 Albemarle County Census data. The figure on the left depicts the distribution of count values for the variable 'birth year'. The two figures on the right depict the most common and least common entries for "birth year".

Figure 6:



The combined figure depicts data from the 1870 Albemarle County Census data. The figure on the left depicts the distribution of count values for the variable 'norm_first_name'. The two figures on the right depict the most common and least common entries for 'norm_first_name'.

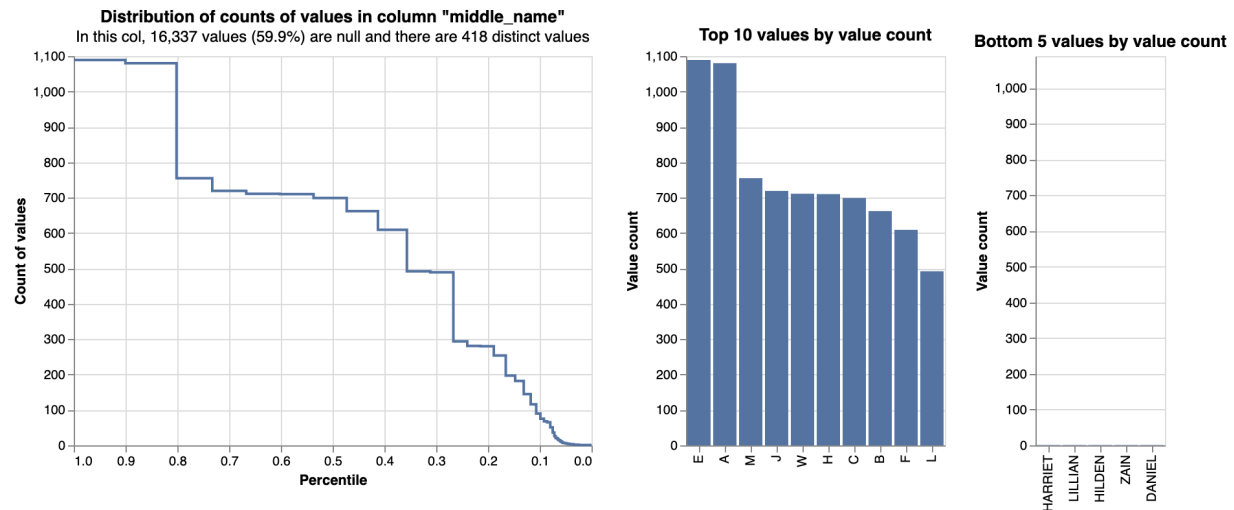
Figure 7:



The combined figure depicts data from the 1870 Albemarle County Census data. The figure on the left depicts the distribution of count values for the variable 'nysiis_last_name'. The two figures on the right depict the most common and least common entries for "nysiis_last_name".

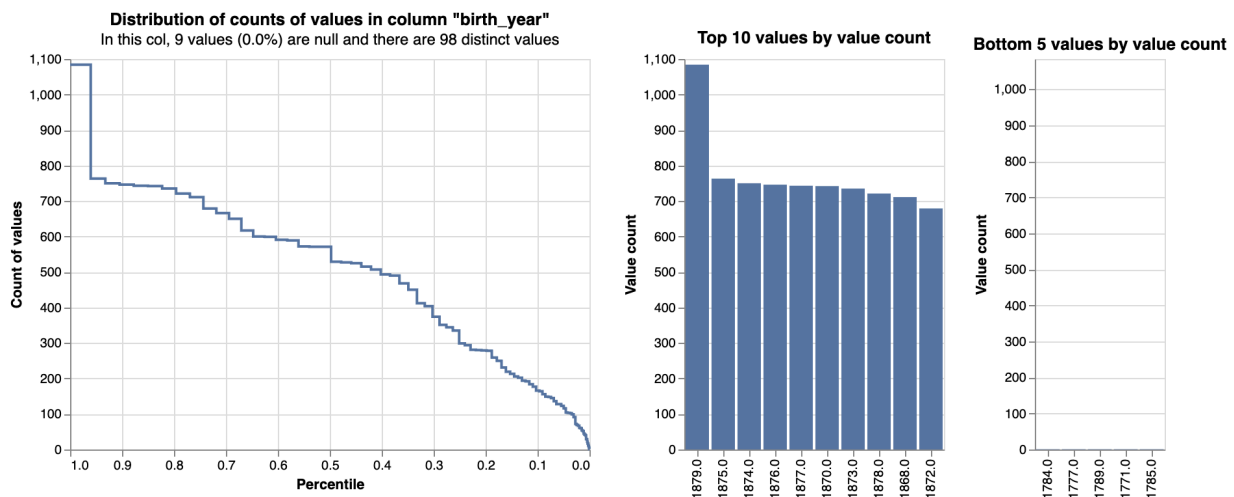
1880 Census Data

Figure 8:



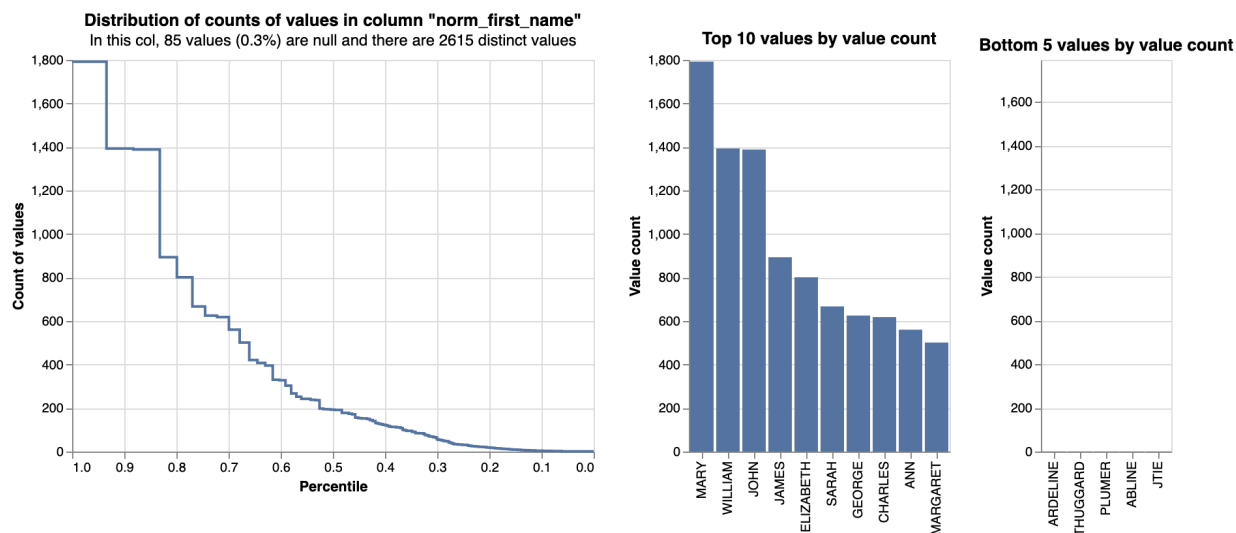
The combined figure depicts data from the 1880 Albemarle County Census data. The figure on the left depicts the distribution of count values for the variable 'middle name'. The two figures on the right depict the most common and least common entries for "middle name". It is apparent that throughout this time period it was more common to simply write down a middle initial rather than record the entire middle name.

Figure 9:



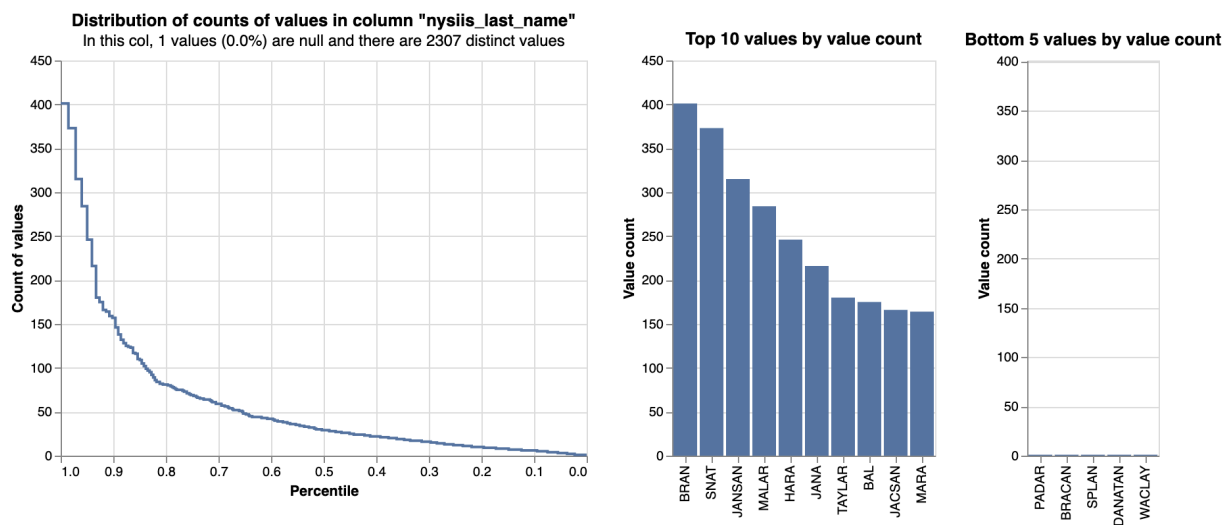
The combined figure depicts data from the 1880 Albemarle County Census data. The figure on the left depicts the distribution of count values for the variable 'birth year'. The two figures on the right depict the most common and least common entries for "birth year".

Figure 10:



The combined figure depicts data from the 1880 Albemarle County Census data. The figure on the left depicts the distribution of count values for the variable 'norm_first_name'. The two figures on the right depict the most common and least common entries for "norm_first_name". The 5 most common names remain the same from 1870 to 1880. 1880 introduced new common first names such as "Sarah" and "Margaret".

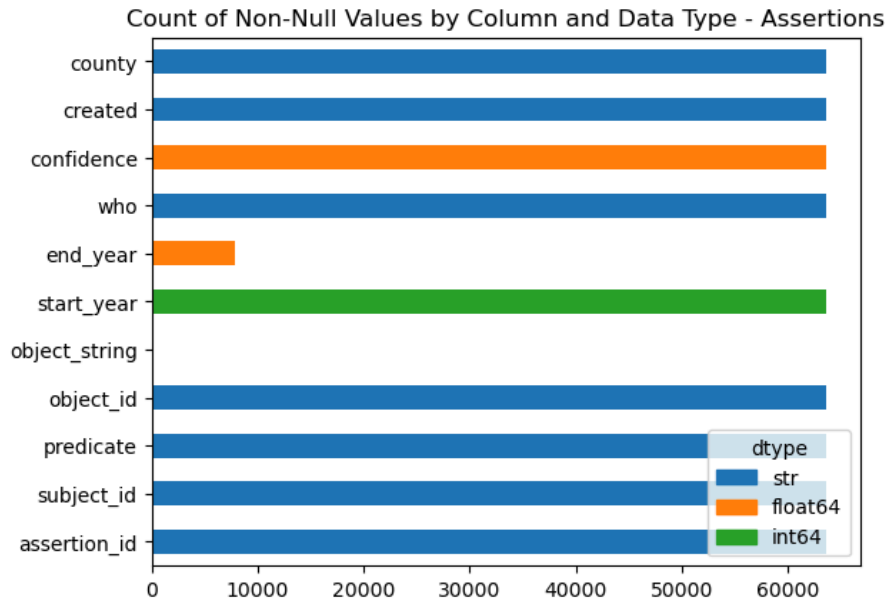
Figure 11:



The combined figure depicts data from the 1880 Albemarle County Census data. The figure on the left depicts the distribution of count values for the variable 'nysiis_last_name'. The two figures on the right depict the most common and least common entries for "nysiis_last_name". The most common last normalized name "Bran" remains the most common from 1870 to 1880. There are some other names that remain in the 10 most common last names such as 'Jansan' and 'Jana'.

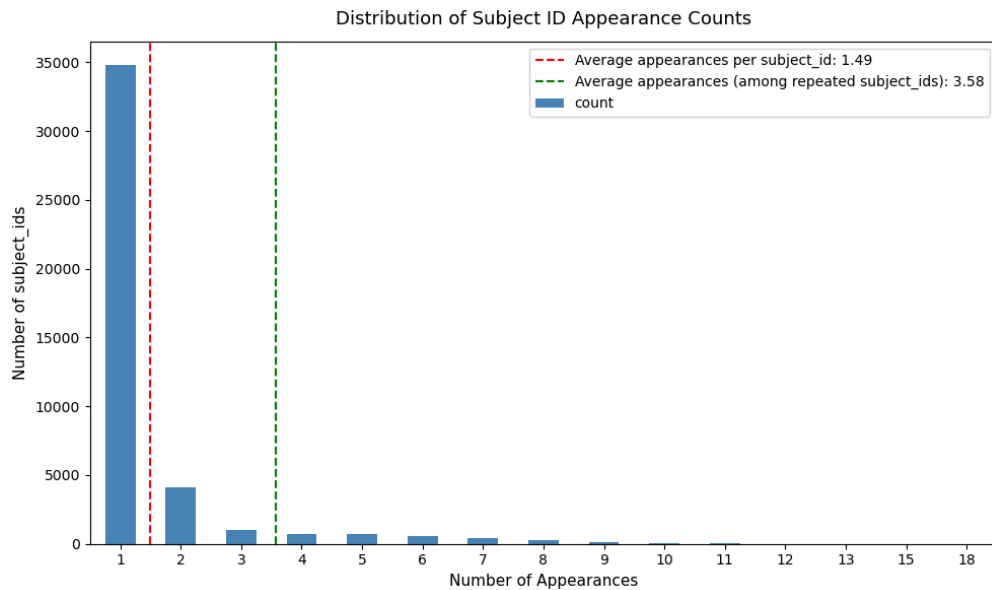
Assertions Data

Figure 12:



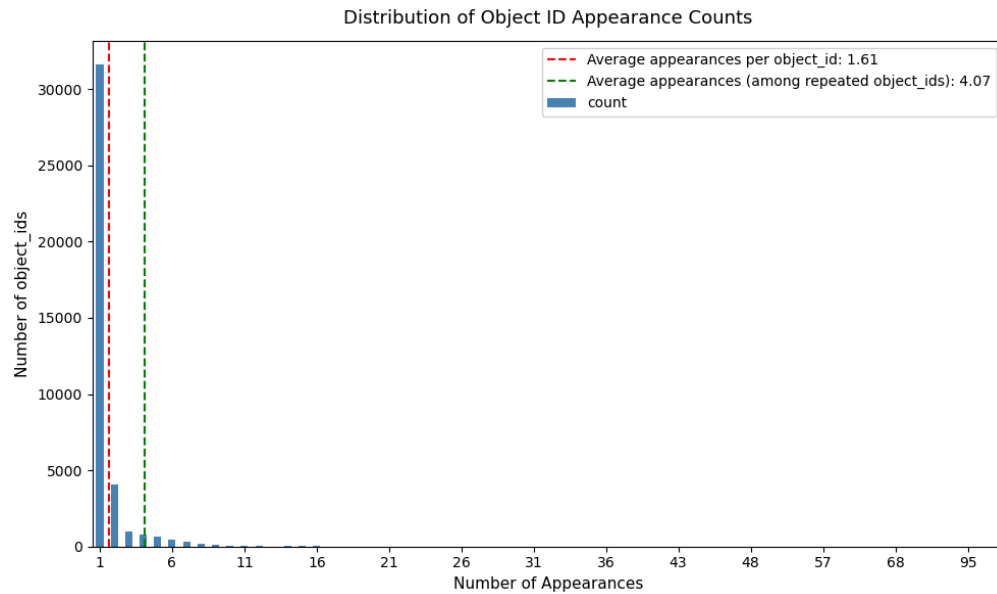
The assertions dataset has **63,636** observations and **11** features/columns. Additionally, the 'object_string' feature has all NA values, and the 'end_year' feature has **55,822** NA values. No other features have NA values.

Figure 13:



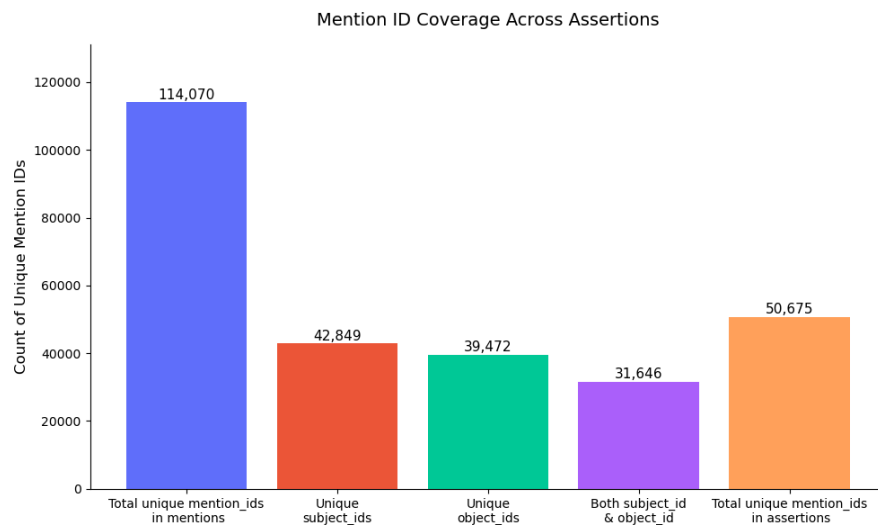
The 'subject_id' feature in the assertions dataset represents one of two individuals in an asserted relationship. On average, each 'subject_id' appears **~1.49** times, and each repeated 'subject_id' appears **~3.58** times.

Figure 14:



The 'object_id' feature in the assertions dataset represents the other individual in an asserted relationship. On average, each 'subject_id' appears **~1.61** times, and each repeated 'subject_id' appears **~4.07** times. Since the “wasEnslavedBy” ‘predicate’ represents a one-sided relationship, some former enslavers appear up to **117** times.

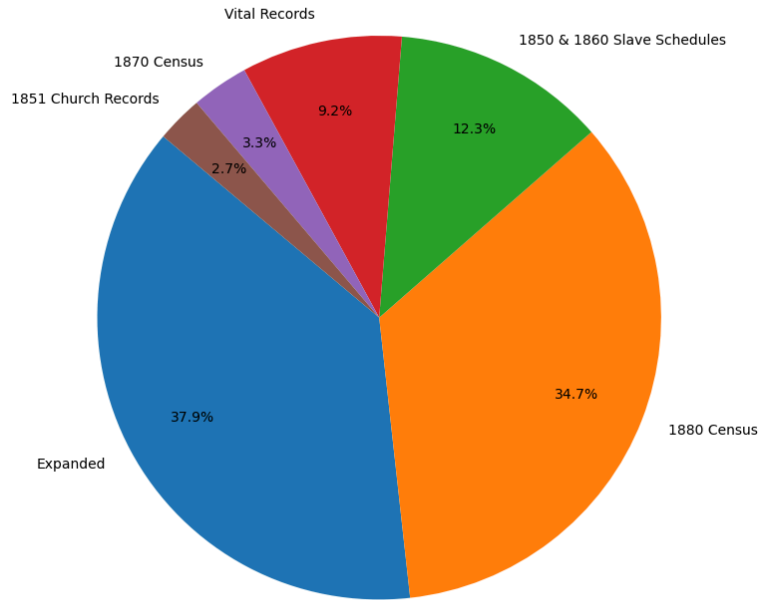
Figure 15:



While there are **114,070** unique mention_ids in the mentions dataset, only **50,675** (~44%) of those individuals have asserted relationships. Moreover, **31,646** (~27.7%) of mention_ids appear as both a 'subject_id' and an 'object_id'.

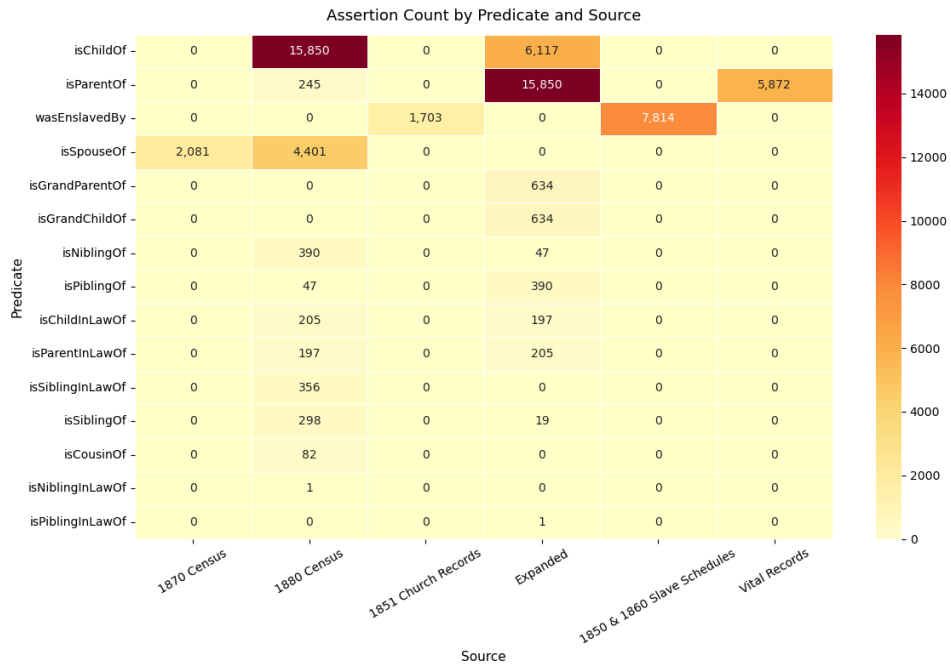
Figure 16:

Distribution of Sources - Assertions



The assertions dataset includes relational assertions from 6 sources. The 'Expanded' category simply represents inverse relationships from any source (e.g., "isChildOf" and "isParentOf" 'predicate' values are both used to represent the same relationship, resulting in two observations per asymmetrical relationship).

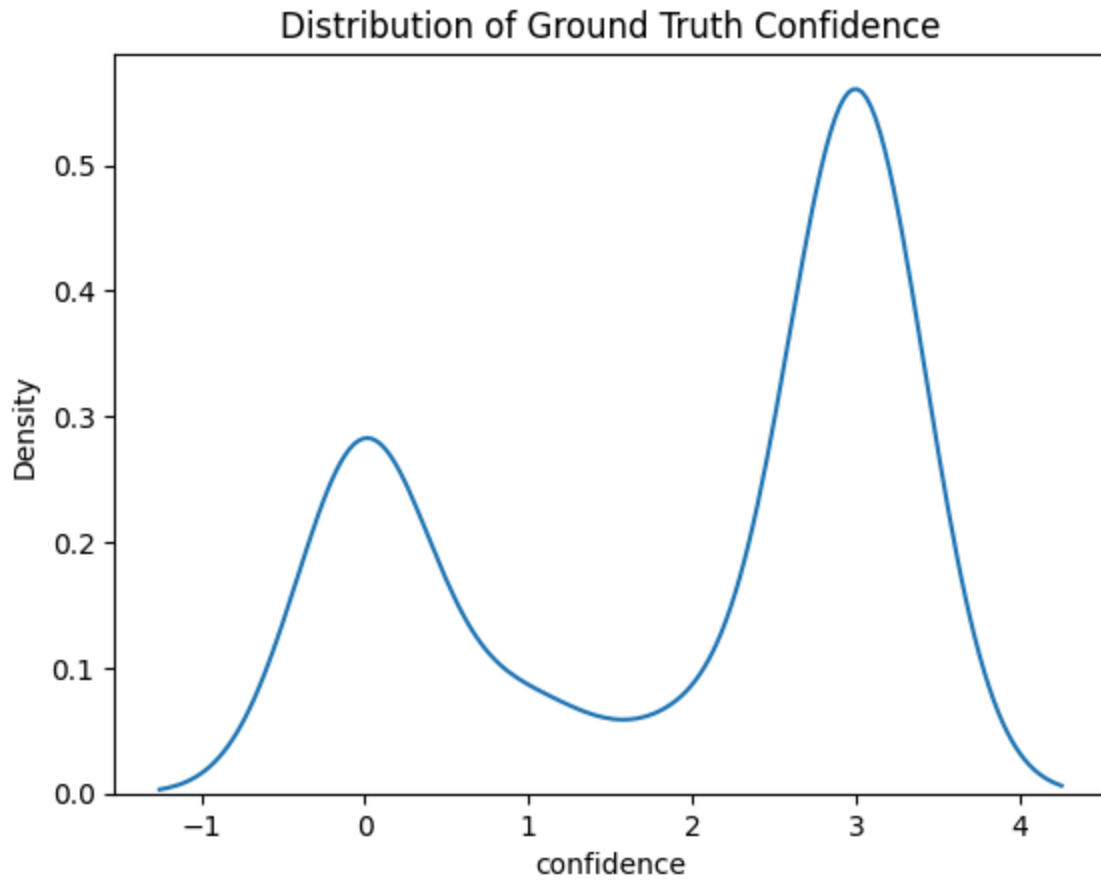
Figure 17:



While there exist 15 ‘predicate’ values (relationship types), the majority of assertions represent parent-child (21,967 observations), former enslaver-former laborer (9,517 observations), and spousal (6,482 observations) relationships.

Ground Truth Data

Figure 18:



The distribution of values for **confidence** within our ground truth dataset is bimodal and left skew, with a moderate peak at 0 and a higher peak at 3. The dataset is imbalanced towards more confident matches.

Analysis of Results

Candidate Matches Data

Figure 19a:

Percent Ground Truth Matched at Match Threshold 0.8

confidence_level	match_percentage
0	0.485
1	0.88
2	1
3	0.995
overall	0.839

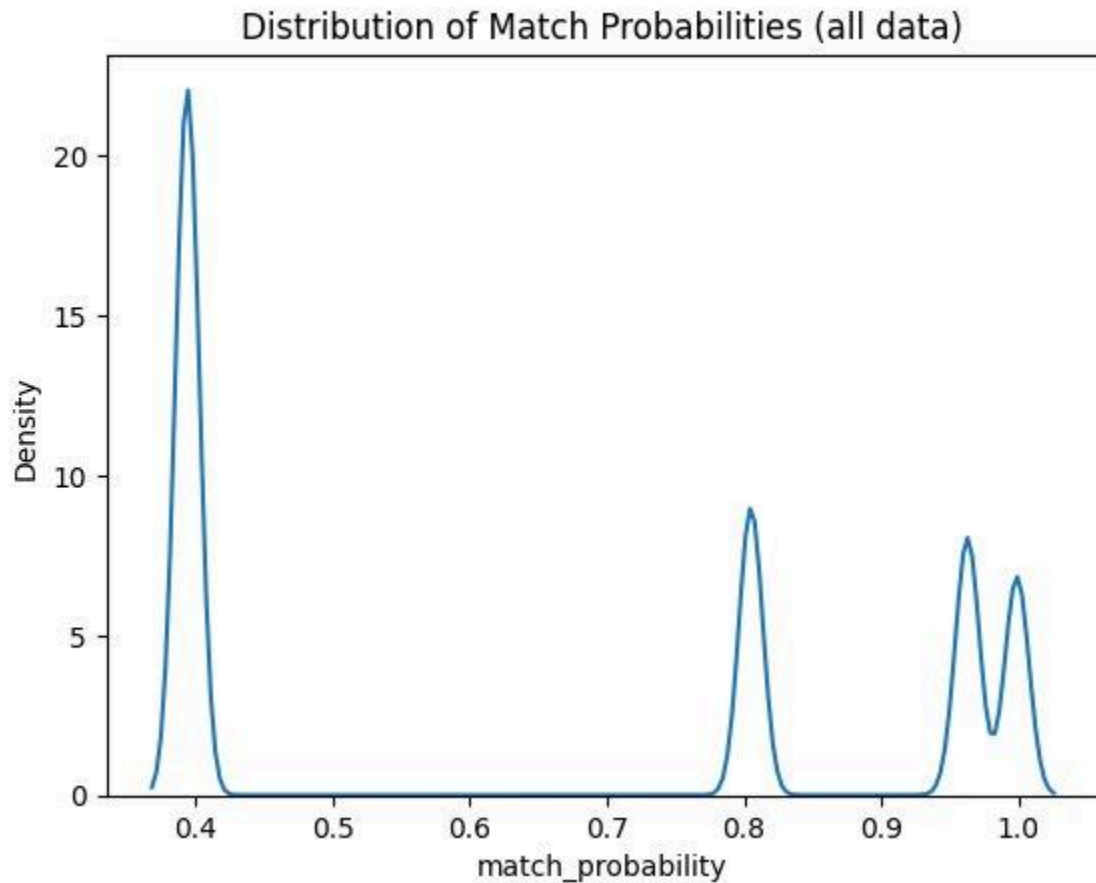
Figure 19b:

Percent Ground Truth Matched at Match Threshold 0.9

confidence_level	match_percentage
0	0.417
1	0.84
2	1
3	0.99
overall	0.813

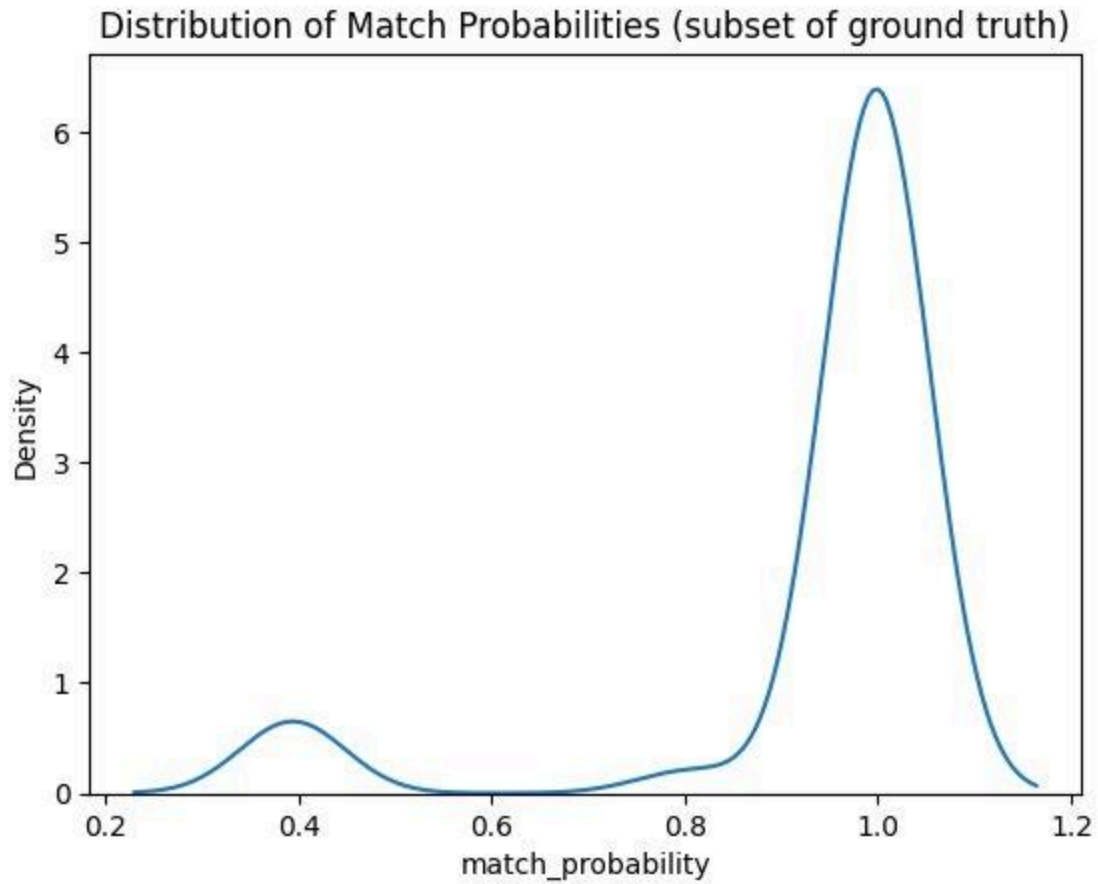
Matched pairs were extracted from predictions as pairs with a math probability greater than 0.8 (10a) and 0.9 (10b). This figure shows what percentage of the ground truth we correctly predicted - overall and at each confidence level.

Figure 20:



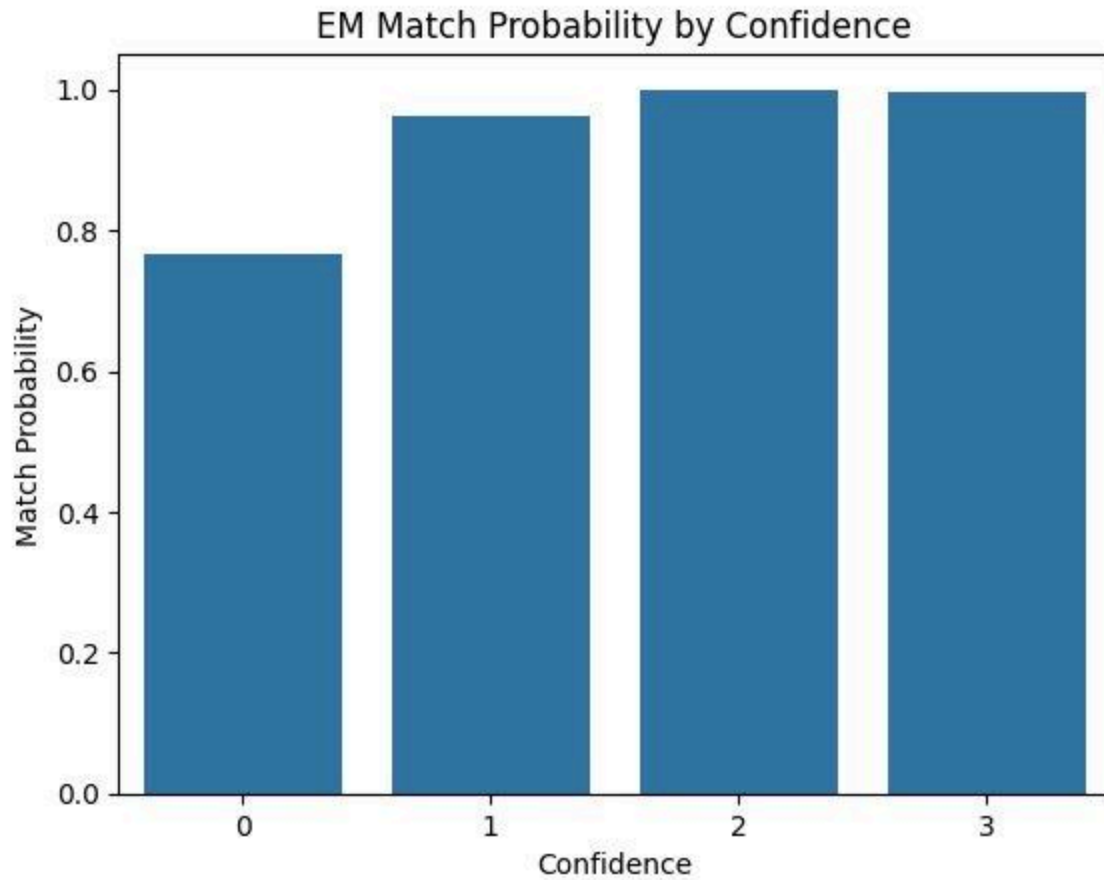
The distribution of our match probabilities is multimodal as we can identify a main peak with a higher density of values at around 0.4, with a moderate peak at around 0.8, and two smaller peaks in the 0.95-1 range.

Figure 21:



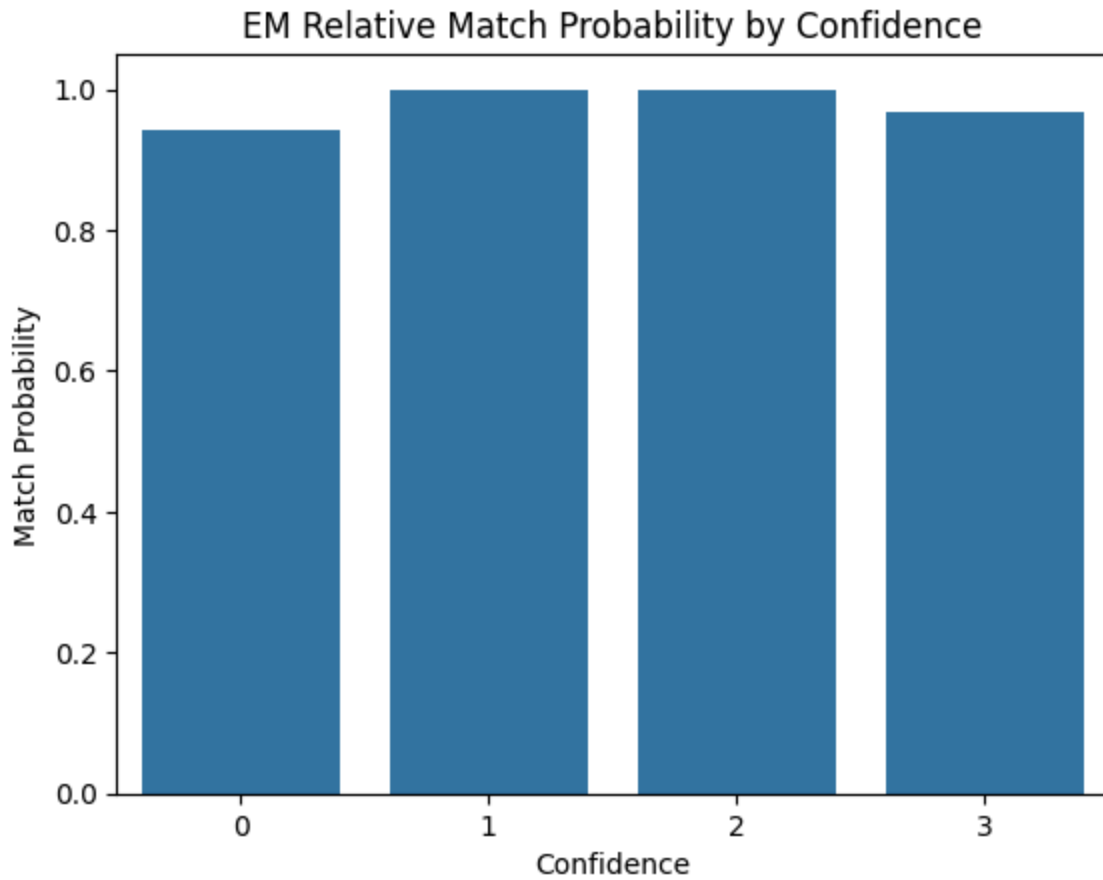
*On the contrary to Figure 11, on the subset of **ground truth** data, we have a strong left-skew distribution - though you could argue bimodality as there is a very small peak at around 0.4. A majority of values are densely packed at around 1 for our EM model match probabilities.*

Figure 22:



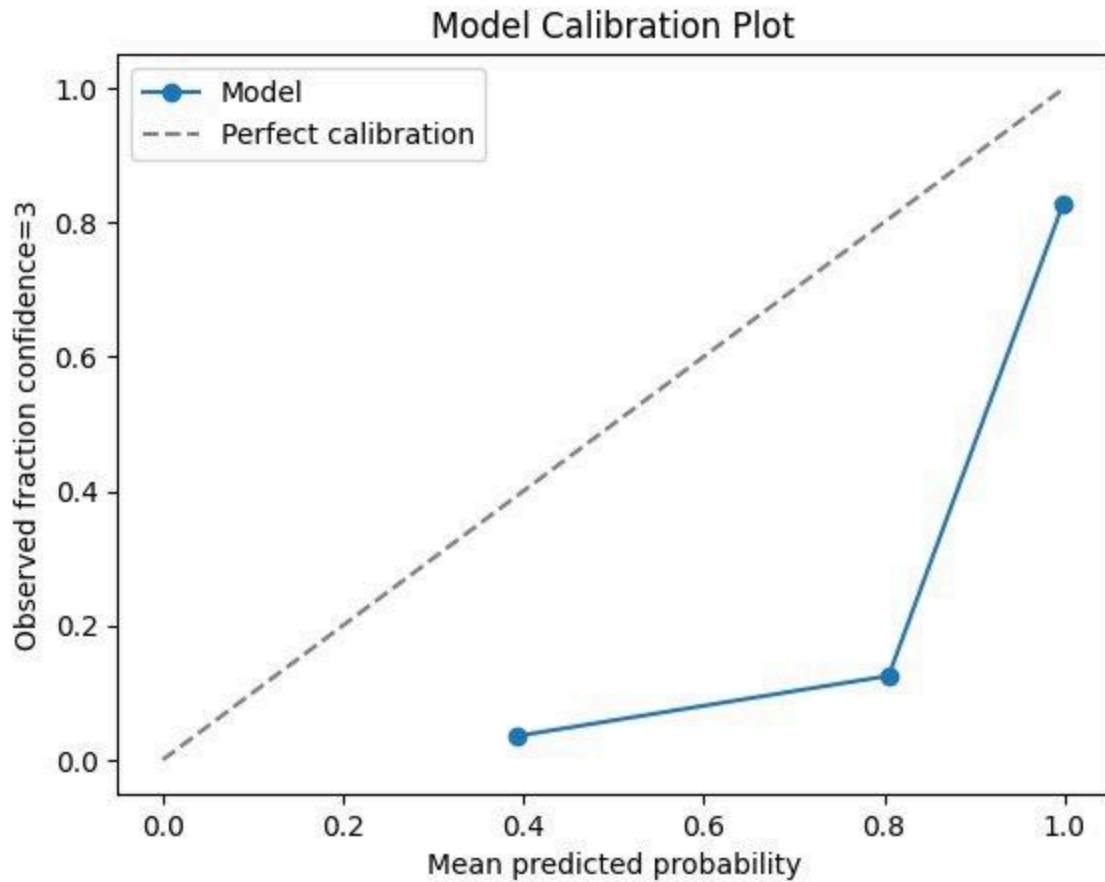
By taking the average match probability for each confidence level, we can see how our model performs at each confidence level. At which we can see a clear pattern of increased confidence with an increased likelihood of a match (based on our EM model).

Figure 23:



*At each grouped confidence level, the average for relative match probability does not appear to follow any pattern, with average values of around 0.9-1 at each confidence level (0-3), meaning that with our **ground truth** data, our relative probabilities fail to provide more confidence for a match.*

Figure 24: Calibration Plot (treating a confidence of 3 as a “perfect match”)



*The area in which our EM model struggles the most with our **ground truth** data is the lack of calibration between predicted probabilities with confidence - treating a confidence of 3 as a “perfect match”. Our highly confident predictions (~1) correspond to a true match rate at ~0.8, and our less confident predictions (0.4) correspond to a true match rate of ~0 with the ground truth data provided.*